INCH-POUND

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MILITARY SPECIFICATION

INTERCOMMUNICATIONS SET, AN/AIC-14A
AUDIO FREQUENCY AMPLIFIER AM-3889/AIC-14A
GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance and acceptance requirements for AM-3889/AIC-14A Audio Frequency Amplifier, herein referred to as the AM-3889.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-5087	Bonding, Electrical and Lighting Protection for Aerospace Systems

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (continued)

MILITARY (continued)

MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base NATO Code Number O-148
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft NATO Code Number H-537
MIL-I-85845	Intercommunication Sets, AN/AIC-14A Master Interphone Control Panel C-2645D/AIC-14A, General Specification for
MIL-I-85846/1	Intercommunication Sets, AN/AIC-14A Microphone Mixing Module
MIL-I-85846/2	Intercommunication Sets, AN/AIC-14A Dual Output Amplifier Module

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of

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STANDARDS (continued)

MILITARY (continued)

MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-965	Parts Control Program
MIL-STD-45662	Calibration Systems Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DOCUMENTS

NAVAL AIR SYSTEMS COMMAND

AD-1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

DL 1794AS9000	Audio Frequency Amplifier, AM-3889/AIC-14A

(Copies of specifications, standards, drawings and documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for the associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet.

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In event of any conflict between the requirements of this specification and specification sheet, the latter shall govern.

3.2 Item description. The AM-3889 is normally used in conjunction with the C-2645D Master Interphone Control Panel (see MIL-I-85845). The AM-3889 shall be capable of accepting two separate microphone inputs. The AM-3889 shall provide amplified voice output signals to headsets at two different crew stations. Two-channel output shall be provided to permit the use of two headsets without interaction or interference. The outputs shall be capable of operating directly into high impedance headsets, low impedance headsets, or loudspeakers without the use of external impedance matching transformers. The AM-3889 is mounted to an aircraft with Mounting Plate, MT-2075/AIC-14. Figure 1 provides an outline drawing of the AM-3889.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.4 Parts and materials.

3.4.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-STD-965, Procedure I.

3.4.2 Parts rescreening. The contractor shall provide 100 percent rescreening of active components such as but not limited to ICs, hybrids, semiconductors and other components, as appropriate to provide a quality product. Active component parts rescreening at the electronic equipment manufacturer's plant shall consist of functional check of critical parameters at +125°C ambient and -54°C to the part specification.

3.4.3 Supplemental requirements. In the event that the contractor proposes to alter the materials or piece parts of the AM-3889 specified herein, the supplemental requirements of 3.4.2.1 through 3.4.2.3 shall be applicable for the selection of parts and materials.

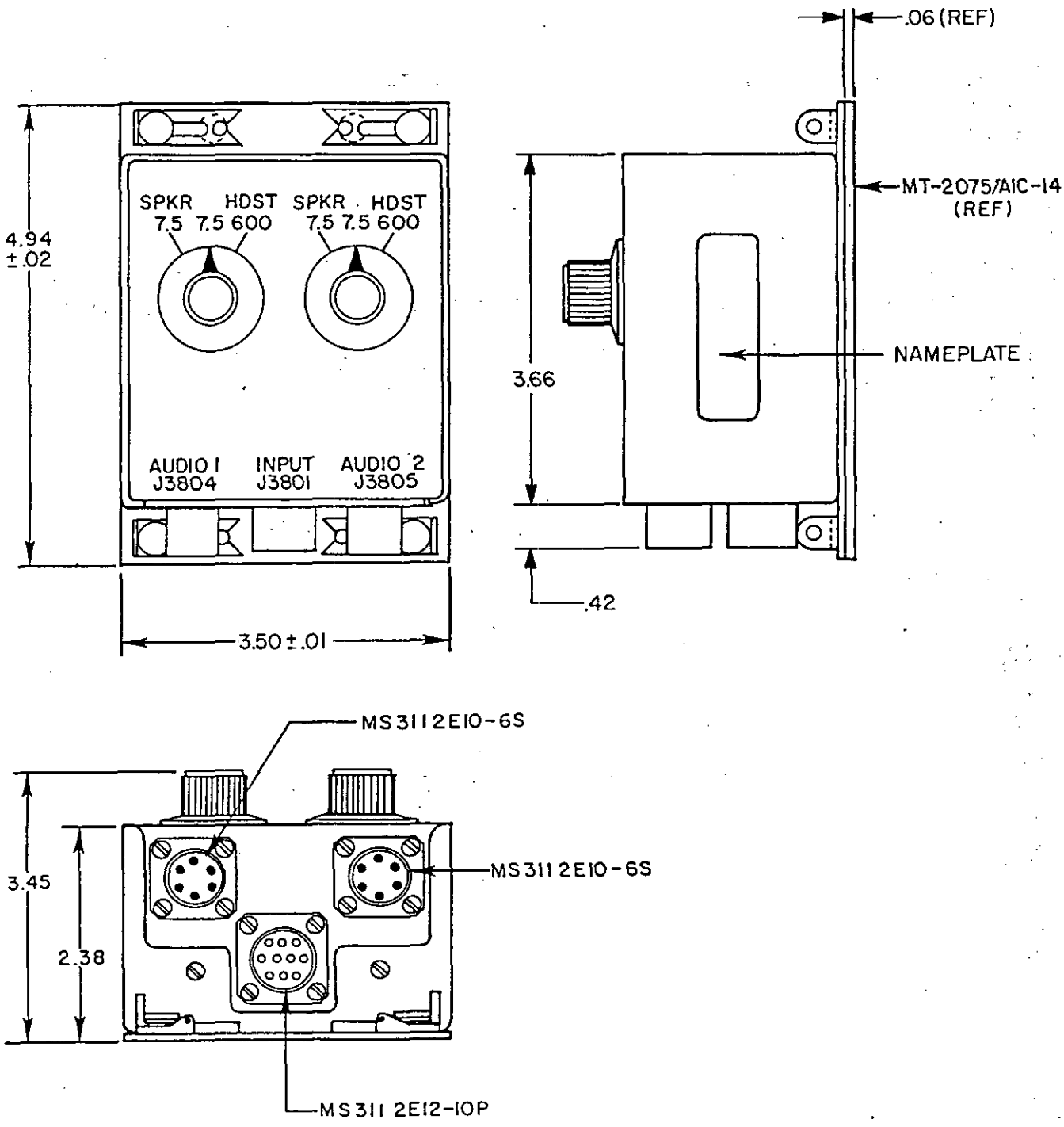
3.4.3.1 Basic conformance. Parts and materials selected for use in the AM-3889 shall be in accordance with MIL-E-5400, unless the parts or materials listed therein will not perform adequately in the intended applications and environments specified herein. When any parts and materials not in accordance with MIL-E-5400 must be used, refer to 3.4.1.

3.4.3.2 Toxic and corrosive fumes. The materials, as used in the manufacture of the AM-3889 and under the conditions therein, shall not liberate deleterious or corrosive fumes. This shall include fungicidal agents that are used.

3.4.3.3 Recycled, virgin and reclaimed materials. Except when intended use of the AM-3889 will be jeopardized by the use of reclaimed or recycled materials, these materials shall be used to the maximum extent possible.

3.5 Design and construction. Design and construction of the AM-3889 shall conform with MIL-E-5400 and DL 1794AS9000 and as specified herein. It

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 1. AM-3889 outline.

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is a condition of final acceptance that the AM-3889 manufactured shall meet all the design, construction and workmanship requirements of this specification and applicable acceptance test documents.

3.5.1 Reliability.

3.5.1.1 Operational stability. The AM-3889 shall operate with required performance, continuously or intermittently, for not less than 1000 hours.

3.5.1.2 Operating life. The AM-3889 shall have a minimum total operating life not less than 10,000 hours when maintenance delineated in the maintenance plan is performed in accordance with the approved procedures.

3.5.1.3 Reliability in mean-time-between-failure (MTBF). The lower test MTBF 0_1 for the AM-3889 shall be not less than 1000 hours.

3.5.2 Interchangeability. Interchangeability shall exist between all AM-3889s and all replaceable assemblies, subassemblies and parts of the AM-3889s in accordance with MIL-E-5400. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance without need for mechanical or electrical matching, alignment or adjustment of parts or circuitry.

3.5.3 Transportability. The AM-3889 shall be transportable by air, rail, truck or ship when packaged as specified in Section 5.

3.5.4 Dimensions. The dimensions of the AM-3889 shall be as shown on Figure 1 and as follows:

a.	Height	4.94 $\pm$ 0.02 inches.
b.	Width	3.50 $\pm$ 0.01 inches.
c.	Depth	3.45 $\pm$ 0.03 inches.

3.5.5 Weight. The weight of the AM-3889 shall be not greater than 2.0 pounds.

3.5.6 Color. The color of the AM-3889 shall be as identified in DL 1794AS9000.

3.5.7 Connectors. The input and output connectors shall be as identified in DL 1794AS9000.

3.5.8 Nameplates and identification marking. Serial number assignment and nameplate approval shall be in accordance with MIL-N-18307. Identification marking shall be in accordance with MIL-STD-130. The following information shall be placed on the nameplate:

a. Unit nomenclature (Audio Frequency Amplifier).

b. Unit designator (AM-3889/AIC-14A).

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c. Contract number.

d. Manufacturer's code number.

e. Equipment drawing number (30003 - 1794AS9000).

f. Technical manual identification.

g. Procuring activity (Department of the Navy - Naval Air Systems Command).

h. Serial number specified in the contract shall be stamped on the AM-3889.

3.5.8.1 Serial numbers. Serial numbers shall be assigned in accordance with MIL-N-18307 for the AM-3889s under contract or purchase order. The contractor shall serialize sequentially all AM-3889s which are delivered under the contract.

3.5.9 Safety. The contractor shall ensure that safety requirements are in accordance with MIL-STD-882.

3.5.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (25 ± 10°C).
b.	Altitude	Normal ground (0 - 5000 feet).
c.	Vibration	None.
d.	Humidity	Room ambient up to 90 percent relative humidity.
e.	Input power	28 volts dc, 20 amperes.

3.5.11 Service conditions. The AM-3889 shall operate as required under any of the environmental service conditions, or any probable combination thereof, as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.5.11.1 Altitude. The AM-3889 shall operate as required at any pressure up to 50,000 feet.

3.5.11.2 Fungus. The AM-3889 shall withstand exposure to fungal growth. Overspraying of the AM-3889 shall be prohibited.

3.5.11.3 Fluids. The AM-3889 shall be unaffected by contact with the following fluids:

a. JP-4 and JP-5 fuels (see MIL-T-5624).

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b. Hydraulic fluids (see MIL-H-83282).

c. Lubricating oil (see MIL-L-7808).

d. Coolants of the fluorocarbon, silicone, silicate ester and glycol families.

3.5.12 Warm-up time. The time required for the AM-3889 to warm up prior to providing voice communications shall be not greater than 10 seconds under standard conditions and 30 seconds under service conditions. The time required for the AM-3889 to warm up in order to meet the performance parameters specified herein shall be not greater than 2 minutes under standard conditions and 4 minutes under service conditions.

3.5.13 Input power. The AM-3889 shall meet all applicable requirements of MIL-STD-704 and shall meet specified performance requirements when energized from a power source having characteristics and limits specified in MIL-STD-704 for 28 volts dc.

3.5.13.1 Steady state power.

3.5.13.1.1 Normal steady state. The AM-3889 shall meet the requirements of MIL-STD-704 for normal steady state dc input power.

3.5.13.1.2 Emergency steady state. The AM-3889 shall meet the requirements of MIL-STD-704 for emergency steady state dc input power.

3.5.13.2 Transients.

3.5.13.2.1 Voltage surge. The AM-3889 shall meet the requirements of MIL-STD-704 for voltage surges. The AM-3889 shall not be failed for degraded performance during the voltage surges if the AM-3889 returns to normal operation without damage after voltage surges pass.

3.5.13.3 Overvoltage, undervoltage and power interruption.

3.5.13.3.1 Overvoltage. The AM-3889 shall meet the requirements of MIL-STD-704 for overvoltage. The AM-3889 shall not be failed for degraded performance during overvoltage exposure if the AM-3889 returns to normal operation without damage after the exposure.

3.5.13.3.2 Undervoltage. The AM-3889 shall meet the requirements of MIL-STD-704 for undervoltage. The AM-3889 shall not be failed for degraded performance during undervoltage exposure if the AM-3889 returns to normal operation without damage after the exposure.

3.5.13.3.3 Power interruption. The AM-3889 shall meet the requirements of MIL-STD-704 for power interruption. The AM-3889 shall not be failed for degraded performance during power interruption of the AM-3889 returns to normal operation without damage after the exposure.

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3.5.14 Electromagnetic interference (EMI). The AM-3889 shall comply with the requirements of MIL-STD-461, Class A1b equipment, as modified herein. These requirements shall be met in all operating modes and at all operating frequencies of the AM-3889 at standard conditions specified in 3.5.10. EMI testing shall be accomplished in accordance with the procedures of MIL-STD-462 to demonstrate compliance with requirements of MIL-STD-461. EMI design guidelines shall be in accordance with AD-1115.

3.5.14.1 Conducted and radiated emissions.

3.5.14.1.1 Conducted emission, power and interconnecting leads. Conducted emissions from the AM-3889 power and interconnecting leads shall be within the limits of MIL-STD-461, Requirement CE03.

3.5.14.1.2 Conducted emissions, power leads, spikes, time domain. Conducted emissions from the AM-3889 power leads shall be within the limits of MIL-STD-461, Requirement CE07.

3.5.14.1.3 Radiated emissions electric field 14 kHz to 10 GHz. Radiated emissions from the AM-3889, electric field, between 14 kHz and 10 GHz, shall be within the limits of MIL-STD-461, Requirement RE02. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized waves.

3.5.14.2 Conducted and radiated susceptibility.

3.5.14.2.1 Conducted susceptibility, power leads, 30 Hz to 50 kHz. The AM-3889 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to electromagnetic energy injected onto its power leads less than or equal to the values of MIL-STD-461, Requirement CS01.

3.5.14.2.2 Conducted susceptibility, power leads, 50 kHz to 400 MHz. The AM-3889 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to 1 volt from a 50-ohm source across a calibrating resistor as specified in MIL-STD-461, Requirement CS02. This requirement shall apply to all neutrals and grounds whether or not they are internally returned to chassis ground.

3.5.14.2.3 Conducted susceptibility, spikes, power leads. The AM-3889 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when spikes each having the waveform specified in MIL-STD-461, Requirement CS06, are applied to the ac and dc power input leads.

3.5.14.2.4 Radiated susceptibility, magnetic induction field, spikes and power frequencies. The AM-3889 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to radiated signals specified in MIL-STD-461, Requirement RS02.

3.5.14.2.5 Radiated susceptibility, electric field, 14 kHz to 40 GHz. The AM-3889 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to the radiated electric fields specified in MIL-STD-461, Requirement RS03, except field levels and modulation shall be as specified in Table I.

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TABLE I. RS03 signal (continuous application test levels).

Frequency (MHz)	Modulation					Power Density (mW/cm^2)		
						Peak 5/	Ave	v/M(Ref) 6/
0.014-850	AM 90% 1 kHz Sine Wave					-	0.66	50
215-225	PAM 1/	20 usec 2/	PW 3/	200	pps 4/	0.126	-	22
215-225	PAM	200 usec	PW	200	pps	0.126	-	22
400-450	PAM	4 usec	PW	300	pps	0.200	-	28
850-940	PAM	10 usec	PW	1000	pps	0.316	-	35
850-940	PAM	125 usec	PW	300	pps	0.316	-	35
940-1215	PAM	10 usec	PW	1000	pps	0.106	-	20
1215-1365	PAM	10 usec	PW	3000	pps	0.106	-	20
1365-2900	PAM	1 usec	PW	1000	pps	0.160	-	25
2900-3100	PAM	3 usec	PW	1000	pps	1.60	-	78
2900-3100	PAM	70 usec	PW	200	pps	1.60	-	78
3100-3500	PAM	10 usec	PW	3000	pps	3.18	-	110
3100-3500	PAM	50 usec	PW	3000	pps	3.18	-	110
3500-9000	PAM	1 usec	PW	1000	pps	0.636	-	50
9000-14000	PAM	0.2 usec	PW	1000	pps	0.106	-	20
14000-18000	PAM	0.2 usec	PW	50000	pps	0.106	-	20
33200	PAM	0.2 usec	PW	2000	pps	0.106	-	20

NOTES:

1/ Pulse amplitude modulated.
2/ Microseconds.
3/ Pulse width.
4/ Pulses per second.
5/ Peak power density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; for example, average power density within the pulse envelope.
6/ Volts per meter equivalent is shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and field strength shown.

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3.5.14.3 Susceptibility interpretation. The criteria of 3.5.14.3.1 and 3.5.14.3.2 shall be applied to determine whether the unit is susceptible.

3.5.14.3.1 Susceptibility to stray magnetic fields. The AM-3889 shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 Hz to 2800 Hz, the properly terminated output from each amplifier circuit shall be not greater than 10.0 millivolts at 400 Hz and shall be not greater than a linear increase of 16 μV/Hz up to 2800 Hz.

3.5.14.3.2 Susceptibility. The AM-3889 shall be designed to resist audio interference including power supply pick-up. The AM-3889 shall exhibit audio outputs not greater than -70 dB within the audio pass band of 100 Hz to 10 kHz when subjected to the radiated and conducted signal exposures required herein.

3.5.14.4 Ground plane interference. The AM-3889 shall be capable of specified performance when subjected to aircraft chassis noise between the interfacing WRAs.

3.5.14.5 Grounding requirements. The grounding system between WRAs of the equipment shall be compatible and consistent with the following aircraft grounding scheme. All WRAs shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, minimize EMI, and to protect personnel from electrical hazards.

3.5.14.6 Bonding. A means of electrically bonding the equipment to the aircraft chassis shall be provided and shall comply with the requirements of MIL-B-5087. Bonding resistance between the bonding surface and all parts of the equipment case shall be not greater than 2.5 milliohms.

3.5.15 Interface.

3.5.15.1 Signal interface. The AM-3889 input and output signals shall have the characteristics specified in Tables II and III.

TABLE II. Input signals.

Signal Name	Characteristics
Microphone No. 1 (MIC No. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Microphone No. 2 (MIC No. 2)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Audio AUDIO	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 27 Vac.
Power (PWR)	+27.5 Vdc $\pm$ 0.5V.

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TABLE III. Output signals.

Signal Name	Characteristics
Speaker No. 1 (SPKR No. 1)	Voice range audio (300 to 7000 Hz) signal ranging from 0 to 4 Vac.
Speaker No. 2 (SPKR No. 2)	Voice range audio (300 to 7000 Hz) signal ranging from 0 to 4 Vac.
Headset (HDST)	Voice range audio (300 to 7000 Hz) signal ranging from 0 to 36 Vac.
Mic Audio (MIC AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging from 0 to 2.0 Vac.

3.5.15.1.1 Grounds.

3.5.15.1.1.1 Power ground (PWR GND). Primary power shall be as specified as power delivered to the AM-3889 by the aircraft generating and distribution system. Primary power returns shall be connected to the chassis of the AM-3889.

3.5.15.1.1.2 System ground (SYS GND). The system ground network shall be established to provide isolation from interference injected onto the power returns. The interconnection of signal returns and signal shield connections form a system ground network. In the AM-3889, system ground shall be isolated from power ground.

3.5.15.1.1.3 Mic return (MIC RTN). The mic return signal shall be isolated from system and power grounds.

3.5.15.1.1.4 Headset return (HDST RTN). The headset return signal shall be to system ground.

3.5.15.2 Operator interface.

3.5.15.2.1 Controls. The AM-3889 shall be provided with operation controls as listed in Table IV.

3.6 Performance. The performance of the AM-3889 shall be as specified in 3.6.1 through 3.6.2.4 and Drawing 1794AS9000. Unless otherwise specified herein, these requirements shall apply under both standard and service conditions (see 3.3.10 and 3.3.11).

3.6.1 Input characteristics. The audio frequency amplifier shall have the input characteristics as specified in 3.6.6.1 through 3.6.1.4.

3.6.1.1 Microphone No. 1 (MIC No. 1). The MIC No. 1 input shall accept signals generated by a microphone or microphone circuit. The input shall have the characteristics as specified in 3.6.1.1.1 through 3.6.1.1.5.

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TABLE IV. Operation controls.

Title	Function
SPKR-HDST	The AM-3889 shall provide two rotary three position switches, each with the following functions: SPKR 7.5 - In the CCW position, the AM-3889 audio output shall be directed to the associated 7.5 ohm speaker with signal power and impedance matching characteristics as specified herein. HDST 7.5 - In the center position, the AM-3889 audio output shall be directed to the associated headset with signal power and impedance matching characteristics as specified herein. HDST 600 - In the CW position, the AM-3889 audio output shall be directed to the associated 600 ohm headset with signal power and impedance matching characteristics as specified herein.

3.6.1.1.1 Microphone current (MIC No. 1). The MIC No. 1 input line shall be capable of supplying 55 + 10 milliamperes to the microphone circuit for carbon microphones or as a source of power to equivalent dynamic microphone and associated preamplifier combinations.

3.6.1.1.2 Signal frequency (MIC No. 1). The AM-3889 shall accept MIC No. 1 input drive signals which range from 300 to 7000 Hz.

3.6.1.1.3 Signal amplitude (MIC No. 1). The AM-3889 shall accept MIC No. 1 input drive signals which range in amplitude from 0.01 to 1.0 Vac.

3.6.1.1.4 Input impedance (MIC No. 1). The impedance of each MIC No. 1 input shall be 100 ohms ± 10 percent.

3.6.1.1.5 Input isolation (MIC No. 1). The isolation between the MIC No. 1 and the MIC No. 2 inputs shall be not less than 20 dB.

3.6.1.2 Microphone No. 2 (MIC No. 2). The MIC No. 2 input shall have characteristics equivalent to those of MIC No. 1 as specified in 3.6.1.1.

3.6.1.3 Audio (AUDIO). The AUDIO input shall accept signals from the interphone lines and have the characteristics specified in 3.6.1.3.1 through 3.6.1.3.3.

3.6.1.3.1 Signal frequency (AUDIO). The AM-3889 shall accept AUDIO input drive signals which range from 300 to 7000 Hz.

3.6.1.3.2 Signal amplitude (AUDIO). The AM-3889 shall accept AUDIO input drive signals which range in amplitude from 0.01 to 27 Vac. The AM-3889 shall be capable of withstanding overvoltage inputs of up to 30 Vac without sustaining damage.

3.6.1.3.3 Input impedance (AUDIO). The impedance of the AUDIO input shall be 600 ohms ± 10 percent.

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3.6.1.4 Power (PWR). The Audio Frequency Amplifier shall accept and operate within the constraints of this specification when supplied with +27.5 + 0.5 Vdc. The maximum load presented by the Audio Frequency Amplifier shall be not greater than 500 milliamperes.

3.6.2 Output characteristics. All signals which are passed through this unit from input to output by being routed only through connectors, wire, printed wiring, switches and relays shall not experience insertion loss in excess of 0.5 dB. The Audio Frequency Amplifier shall have the output characteristics specified in 3.6.2.1 through 3.6.2.5.4.

3.6.2.1 Speaker No. 1 (SPKR No. 1). The SPKR No. 1 output shall have the characteristics specified in 3.6.2.1.1 through 3.6.2.1.6.

3.6.2.1.1 Signal frequency (SPKR No. 1). The SPKR No. 1 output shall be in accordance with the response curves shown on Figure 2 when terminated in a 7.5 ohm load.

3.6.2.1.2 Signal amplitude/gain (SPKR No. 1). The output voltage shall be 11 + 2 dB below the input for speaker operation. The input/output characteristic (gain) shall be linear within ±1 dB for input levels between 0 and 15.0 volts.

3.6.2.1.3 Distortion (SPKR No. 1). The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than the limits established in Table V.

TABLE V. Distortion limits.

Environmental Condition	Output Level (watts)	Distortion Limit (percent)
Standard	2	5
All environmental	2	10

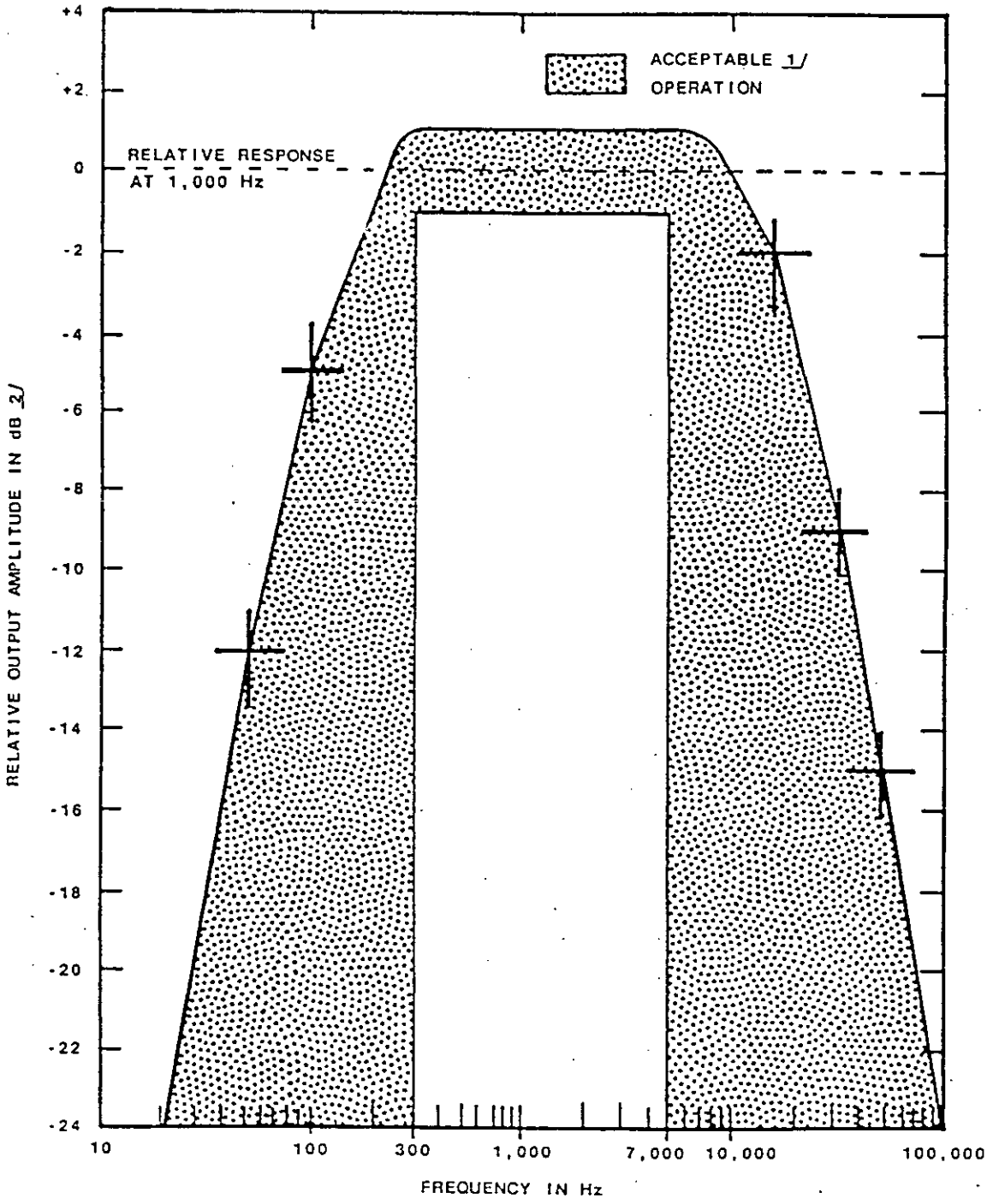
Where:
Standard - Standard conditions specified in MIL-T-5422.
All - Environmental envelope specified in MIL-E-5400 for Class II equipment.

3.6.2.1.4 Output power (SPKR No. 1). The output of the amplifier shall be capable of delivering 2 watts into a 7.5 ohm resistive load.

3.6.2.1.5 Amplifier stability (SPKR No. 1). The amplifier shall operate with no evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short or complex impedance terminations. The amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

3.6.2.1.6 Internal noise (SPKR No. 1). The amplifier shall have an output not greater than 5 millivolts throughout the environmental envelope in MIL-E-5400 for Class II equipment when operated with the input shorted.

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1/ Termination into 600 ohms or 7.5 ohms.
2/ Output amplitude is measured with respect to the amplitude at 1000 Hz.

FIGURE 2. Audio frequency response.

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3.6.2.2 Speaker No. 2 (SPKR No. 2). The SPKR No. 2 output shall have characteristics equivalent to those of SPKR No. 1 as specified in 3.6.2.1.

3.6.2.3 Headset (HDST). The AM-3889 shall provide two HDST outputs. Both outputs shall be capable of operating with 600 ohm (HIGH) or 7.5 ohm (LO) headsets. Selection of headset impedance shall be provided by a switch on the front of the unit. Each HDST output shall have characteristics which comply with the requirements of 3.6.2.3.1 through 3.6.2.3.6.

3.6.2.3.1 Signal frequency (HDST). The HDST output amplitude shall be in accordance with the response curves shown on Figure 2 when terminated by either 7.5 or 600 ohm loads.

3.6.2.3.2 Signal amplitude/gain (HDST). The output voltage for a HDST HIGH selection shall be 0 + 2 dB above the input. The output voltage for a HDST LO selection shall be 17.6 + 2 dB below the input. The input/output characteristic (gain) shall be linear within ±1 dB for input levels between 0 and 15.0 volts rms.

3.6.2.3.3 Distortion (HDST). The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 5 percent for standard conditions as specified by Section 3.1a of MIL-STD-810 at 300 milliwatts output and 8 percent through the environmental envelope specified herein.

3.6.2.3.4 Output power (HDST). The output of the amplifier shall be capable of delivering 2 watts into a 600 ohm resistive load when the HDST 600 position has been selected on the SPKR-HDST control or when HDST 7.5 has been selected on the SPKR-HDST control and the output is terminated with a 7.5 ohm noninductive load.

3.6.2.3.5 Amplifier stability (HDST). The amplifier shall operate with no evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short or complex impedance terminations. The amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

3.6.2.3.6 Internal noise (HDST). The amplifier shall have an output of less than 5 millivolts throughout the environmental envelope specified in MIL-E-5400 for Class II equipment when operated with the input shorted.

3.6.2.4 Microphone current. A filtered current source to power two microphones shall be capable of providing 110 ± 20 ma.

3.6.2.5 Microphone audio out. The MIC AUDIO OUT output shall have the characteristics specified in 3.6.2.5.1 through 3.6.2.5.4.

3.6.2.5.1 Signal frequency (MIC AUDIO OUT). The output shall be not greater than +2 dB from the 1000 Hz output value, over the range of frequencies from 300 Hz to 7000 Hz.

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3.6.2.5.2 Signal amplitude (MIC AUDIO OUT). The output voltage shall be 0 ± 1 dB from an input signal of 1 volt at 1000 Hz.

3.6.2.5.3 Distortion (MIC AUDIO OUT). The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 5 percent.

3.6.2.5.4 Power out (MIC AUDIO OUT). The unit shall be capable of delivering an output power of 20 mW when properly terminated with a 100 ohm load.

3.7 Shop repairable assemblies (SRA). The AM-3889 shall contain the SRAs listed in Table VI.

TABLE VI. Shop repairable assemblies.

Title	Specification
Microphone Mixing Module	MIL-I-85846/1
Dual Output Amplifier	MIL-I-85846/2

3.7.1 Microphone mixing module. The microphone mixing module provides amplification to two independent microphone audio signals and combines these two signals to provide a single output. The output shall be capable of supplying 20 milliwatts into a 100 ohm load. The module also contains a microphone current source with a low pass filter for attenuation of ripple. Performance shall be in accordance with MIL-I-85846/1.

3.7.2 Dual output amplifier. The dual output amplifier provides two power amplifiers whose inputs are driven in parallel and whose outputs are capable of being independently switched to separate output loads. The outputs shall each be capable of supplying at least 2 watts into a 7.5 ohm load. Performance shall be in accordance with MIL-I-85846/2.

3.8 Workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9 and the following:

a. Printed wiring board coatings shall be free of dirt and metal particles.

b. Chemical coated and painted surfaces shall have uniformly coated surfaces.

c. Metal parts shall be free of burrs or sharp edges.

4. QUALITY ASSURANCE PRODUCTS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment and inspection facility. The contractor shall ensure that inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of the required inspections. Inspection equipment used in the inspections specified herein shall be maintained in calibration by a facility acceptable to the procuring activity which meets the requirements of MIL-STD-45662.

4.1.3 Quality assurance terms. Quality assurance terms used in this specification shall be as specified in MIL-STD-109.

4.1.4 Classification of inspections. The inspection requirements specified herein are classified as:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

c. Special inspections (see 4.4).

4.1.5 Inspection conditions. Unless otherwise specified in the contract, all inspections shall be performed in accordance with the test conditions specified in 3.5.10.

4.2 First article inspection. First article inspections specified in 3.3 shall be conducted by the contractor on not less than one AM-3889 representative of the processes, methods and techniques used on the AM-3889s under contract. The first article inspection shall include all inspections required to verify compliance with all electrical and environmental requirements of this specification. The Government quality assurance representative and the procuring activity shall be advised as to when inspections are to be conducted so as to allow monitoring if desired. Contractors not having adequate facilities to conduct all the required inspections may obtain the services of a commercial inspection laboratory only when said laboratory is acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that the AM-3889s conform to specified requirements. First article inspection shall include the following tests:

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a. Altitude (see 4.2.1.1).

b. High temperature (see 4.2.1.2).

c. Low temperature (see 4.2.1.3).

d. Humidity (see 4.2.1.4).

e. Salt fog (see 4.2.1.5).

f. Vibration (see 4.2.1.6).

g. Mechanical shock (see 4.2.1.7).

h. Electromagnetic interference (see 4.2.2).

i. Electrical requirements inspection (see 4.2.3).

4.2.1 Environmental tests. Environmental tests shall be in accordance with MIL-STD-810, except as modified herein. The AM-3889 shall operate as required herein, during (unless as otherwise indicated) and after each environmental test. The AM-3889 shall not sustain either physical damage or performance degradation during or as a result of exposure to the tests. The AM-3889 shall be allowed a ±3 dB gain (see 3.6.2.1.2 and 3.6.2.3.2) when subjected to hot/cold temperature cycling.

4.2.1.1 Altitude. The AM-3889 shall be subjected to an altitude test in accordance with MIL-STD-810, Method 500.2, Procedure II. The test altitude shall be limited to 50,000 feet and the rate of change shall be not greater than 2000 feet per minute.

4.2.1.2 High temperature. The AM-3889 shall be subjected to high temperature operating test in accordance with MIL-STD-810, Method 501.2, Procedure II, except steps 3 through 6 of the procedure shall be deleted. The chamber temperature for step 7 shall be +71°C.

4.2.1.3 Low temperature. The AM-3889 shall be subjected to low temperature operating tests in accordance with MIL-STD-810, Method 502.2, Procedure II. The chamber temperature shall be -54°C.

4.2.1.4 Humidity. The AM-3889 shall be subjected to humidity test in accordance with MIL-STD-810, Method 507.2, Procedure III, Figure 507.2-3. The test duration shall be 10 cycles.

4.2.1.5 Salt fog. The AM-3889 shall be subjected to salt fog test in accordance with MIL-STD-810, Method 509.2, Procedure I. The test duration shall be 48 hours. Following the 48-hour drying period, the AM-3889 shall meet performance requirements specified herein.

4.2.1.6 Vibration. The AM-3889 shall be subjected to random vibration exposure in accordance with MIL-STD-810, Method 514.3, Procedure I for Category 5 equipment. Both functional and endurance tests shall be conducted using the vibration spectra shown on Figure 3. Vibration shall be applied

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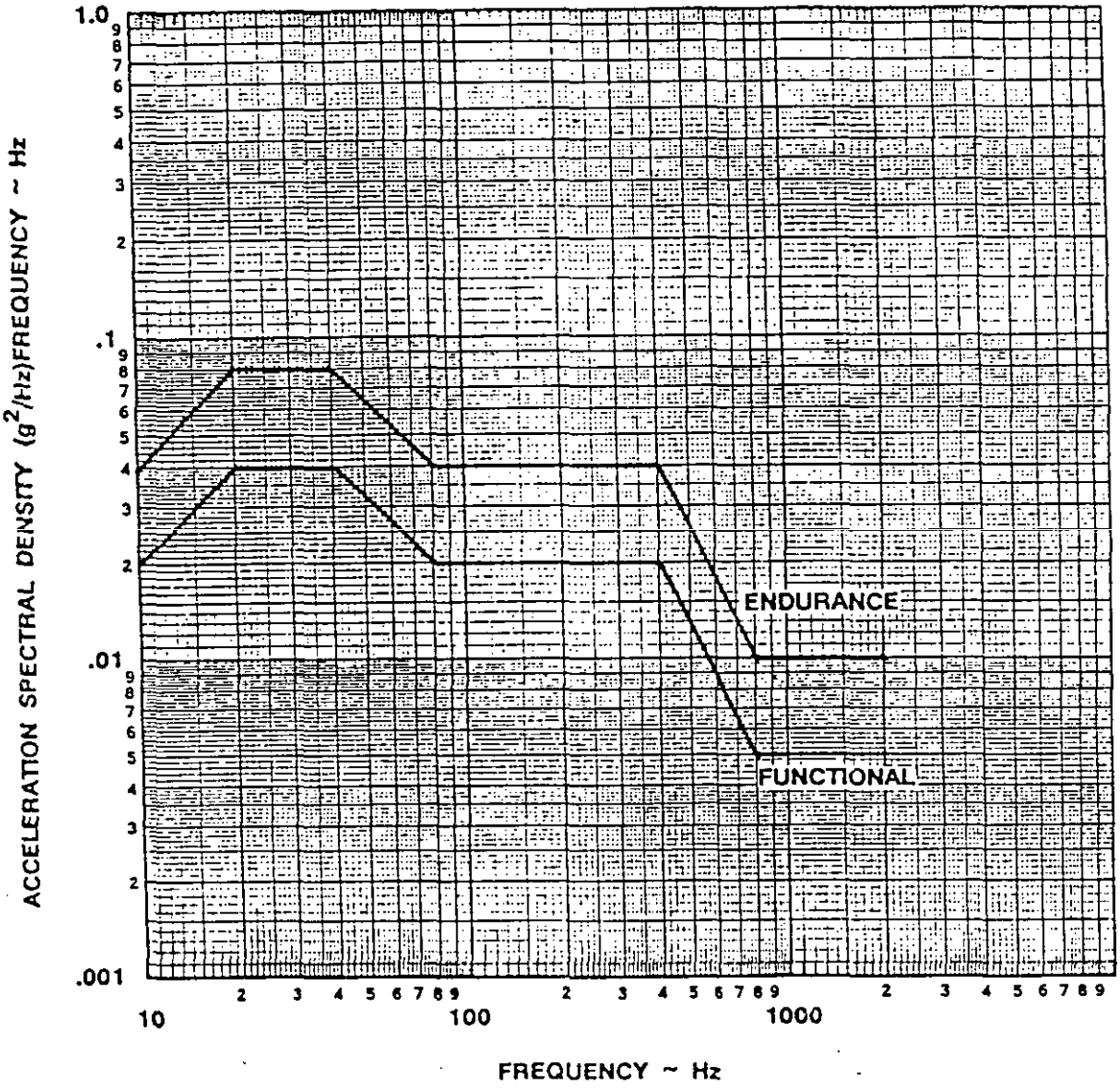


FIGURE 3. Vibration test spectrum.

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sequentially along each of three mutually perpendicular axes for one-half hour at the functional level, and one hour at the endurance level followed by one-half hour at the functional level. The AM-3889 shall operate as required during exposure to the functional level and immediately before and after exposure to the endurance level.

4.2.1.7 Shock. The AM-3889 shall be subjected to shock tests in accordance with MIL-STD-810.

4.2.1.7.1 Functional. The AM-3889 shall withstand three shocks in each direction along three mutually perpendicular axes in accordance with MIL-STD-810, Method 516.3, Procedure I and Figure 516.3-4 for operational test.

4.2.1.7.2 Crash hazard. The AM-3889 shall be subjected to two shocks in each direction along three mutually perpendicular axes in accordance with MIL-STD-810, Method 516.3, Procedure V and Figure 516.3-4 for crash hazard test. There shall be no failure of the mounting attachment and the AM-3889 shall remain in place and not create a hazard. Bending and distortion shall be permitted.

4.2.1.7.3 Bench handling. The AM-3889 shall be tested in accordance with MIL-STD-810, Method 516.3, Procedure VI.

4.2.2 Electromagnetic interference (EMI). The AM-3889 shall be subjected to the EMI tests specified in 4.2.2.1 through 4.2.2.6 in accordance with the test methods of MIL-STD-462 to verify that AM-3889 meets the requirements of MIL-STD-461, Class A1, Category A1b, for the EMI tests herein.

4.2.2.1 Conducted and radiated emission.

4.2.2.1.1 Conducted emissions, power and interconnecting leads, 0.15 to 50 MHz. Conducted emissions measurements shall be made over the frequency range of 0.15 to 50 MHz on power leads in accordance with MIL-STD-462, Test Method CE03, and interconnecting leads in accordance with MIL-STD-462, Test Method CE04.

4.2.2.1.2 Conducted emissions power leads, spikes, time domain. The contractor shall devise a test to demonstrate compliance with the requirements of 3.5.14.1.3.

4.2.2.1.3 Radiated emissions, magnetic field, 0.03 to 50 kHz. Radiated emissions emanating from the AM-3889 shall be measured over frequency range of 0.03 to 50 kHz (magnetic field) in accordance with MIL-STD-462, Test Method RE02.

4.2.2.2 Conducted and radiated susceptibility.

4.2.2.2.1 Conducted susceptibility, power leads, 30 Hz to 50 kHz. The AM-3889 shall be subjected to conducted susceptibility tests by having signals over the range of 30 Hz to 50 kHz injected on the power leads in accordance with MIL-STD-462, Test Method CS01. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

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4.2.2.2.2 Conducted susceptibility, power leads, 50 kHz to 400 MHz. The AM-3889 shall be subjected to conducted susceptibility tests by having signals over the range of 50 kHz to 400 MHz injected on the power leads in accordance with MIL-STD-462, Test Method CS02. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.3 Conducted susceptibility, spikes, power leads. The AM-3889 shall be subjected to conducted susceptibility tests consisting of superimposing a 32-volt maximum spike on the power line voltage waveform in accordance with MIL-STD-462, Test Method CS06. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.4 Radiated susceptibility magnetic induction field spikes and power frequencies. The AM-3889 shall be subjected to radiated susceptibility tests consisting of spikes and power frequencies in accordance with MIL-STD-462, Test Method RS02. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.1.

4.2.2.2.5 Radiated susceptibility, electric field, 14 kHz to 40 GHz. The AM-3889 shall be subjected to radiated susceptibility tests over the frequency range specified in Table I, electric field, in accordance with MIL-STD-462, Test Method RS03.

4.2.2.2.6 Ground plane interference. The contractor shall devise a test to demonstrate compliance with the requirements of 3.5.14.4. This test shall be included as a part of the EMI test plan.

4.2.3 Electrical requirements inspection. The contractor shall conduct inspections necessary to demonstrate compliance to the electrical requirements of Section 3 and DL 1794AS9000. Demonstrations shall include but not be limited to the tests herein.

4.2.3.1 Warm-up time. Time required for the AM-3889 to stabilize after being energized at standard and environmental conditions shall be monitored to assure that the AM-3889 meets the requirements of 3.5.12.

4.2.3.2 Input power.

4.2.3.2.1 Steady state power.

4.2.3.2.1.1 Normal steady state. The AM-3889 shall be checked operationally with input voltage set at 22 volts dc and 29 volts dc in accordance with MIL-STD-704 for normal steady state test. The AM-3889 shall be operated long enough to assure proper operation with each input voltage.

4.2.3.2.1.2 Emergency steady state. The AM-3889 shall be checked operationally with input voltage set at 16 volts dc and 29 volts dc in accordance with MIL-STD-704 for emergency steady state test. The AM-3889 shall be operated long enough to assure proper operation with each input voltage.

4.2.3.2.2 Transient testing.

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4.2.3.2.2.1 Voltage surge. The AM-3889 shall be subjected to input voltage surges of 18 to 50 volts dc for 0.013 to 0.042 second (transient duration) in accordance with MIL-STD-704 for transient test. The AM-3889 operation shall be observed for permanent degradation.

4.2.3.2.3 Overvoltage, undervoltage and power interruption.

4.2.3.2.3.1 Overvoltage. The AM-3889 shall be subjected to input voltage of 50 volts dc for 0.045 second in accordance with MIL-STD-704 for overvoltage test. The AM-3889 operation shall be observed for permanent degradation.

4.2.3.2.3.2 Undervoltage. The AM-3889 shall be subjected to a change of input voltage from normal to 0 volt dc for 5 seconds in accordance with MIL-STD-704 for undervoltage test. The AM-3889 operation shall be observed for permanent degradation.

4.2.3.2.3.3 Power interruption. The AM-3889 shall be subjected to input power interruption of 50 ms in accordance with MIL-STD-704 for power interruption test. The AM-3889 operation shall be observed for permanent degradation.

4.3 Quality conformance inspection.

4.3.1 Individual inspections. Each production AM-3889 submitted for acceptance shall be subjected to the required inspections specified herein. These inspections shall be adequate to determine compliance with the requirements for material, workmanship, operational performance and environmental stress screening (ESS).

a. Examination of product (see 4.3.1.1).

b. Operational inspection (see 4.3.1.2).

c. Environmental stress screening (ESS) (see 4.3.1.3).

d. Production reliability acceptance test (PRAT) (see 4.3.1.4).

4.3.1.1 Examination of product. Each production AM-3889 shall be examined carefully to ensure compliance with requirements of workmanship, marking, interchangeability, design and construction.

4.3.1.2 Operational inspections. Each production AM-3889 shall be subjected to an operational inspection. The inspection shall demonstrate and test all modes of operation; all input/output signals shall be tested and the results recorded. During the inspection sufficient characteristics shall be measured and recorded to assure that the AM-3889s submitted for acceptance meet all the requirements of this specification.

4.3.1.2.1 Input impedance. Verify that the input impedances are in accordance with 3.6.1.1.3, 3.6.1.2 and 3.6.1.3.3 herein. Impedances shall be measured at 1000 Hz. The input impedance shall be measured using the resistance substitution method.

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4.3.1.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.6.2.1.2 and 3.6.2.3.2.

4.3.1.2.3 Frequency response. Holding the input signal amplitude constant, vary the frequency between 300-7000 Hz. Verify that the amplifier gain meets the requirements of 3.6.2.1.1 and 3.6.2.3.1.

4.3.1.2.4 Linearity. Using a 1000 Hz input signal which is varied between 0.01 Vac and 15 Vac, verify that the amplifier meets the requirements of 3.6.2.1.2.

4.3.1.2.5 Output power. With the line out terminated in accordance with 3.6.2.3.4 or 3.6.2.1.4, apply a 1000 Hz signal of appropriate amplitude to verify the requirements of 3.6.2.3.4, 3.6.2.1.4 and 3.6.2.4.

4.3.1.2.6 Distortion. Verify the requirements of 3.6.2.1.3 and 3.6.2.3.3.

4.3.1.2.7 Stability. Verify the requirements of 3.6.2.1.5, 3.6.2.3.5 and 3.6.2.4.

4.3.1.2.8 Internal noise. Verify the requirements of 3.6.2.1.6, 3.6.2.3.6 and 3.6.2.4.

4.3.1.2.9 Controls. Verify control operation of 3.5.15.2.

4.3.1.2.10 Susceptibility to stray magnetic fields. Verify the performance of 3.5.14.2.4.

4.3.1.3 Environmental stress screening (burn-in). An environmental stress screening inspection shall be required on all production AM-3889s. Prior to and after each submission to the screening inspection a full electrical performance inspection under standard conditions shall be performed on each production AM-3889. Each production AM-3889 shall be subjected to a workmanship random vibration of 6.3 g's rms with a frequency of 20 to 2000 Hz, as measured on the vibration fixture at the mounting points (see Figure 4). The vibration duration shall be not less than 10 minutes (failure free) in the axis that is perpendicular to the majority of the printed wiring boards. The production AM-3889 shall be mounted to the vibration fixture by its normal mounting means so as to meet the vibration requirements specified herein. The production AM-3889 shall be continuously energized and electrically monitored for the duration of the inspection.

After the vibration inspection, each production AM-3889 shall be subjected to not less than 72 hours of thermal stress screening, of which not less than 45 hours shall be with the AM-3889 energized. The thermal stress screening inspection shall consist of nine 8-hour temperature cycles of which each cycle contains a 3-hour cold period at -54°C (AM-3889 deenergized) and a 5-hour hot period at +71°C (AM-3889 energized). Each production AM-3889 shall be de-energized for the duration of the cold period except for the time needed to perform the electrical performance inspection specified herein (see Figure 5). Each production AM-3889 shall be energized for the full duration of the hot period. An electrical performance inspection shall be performed once

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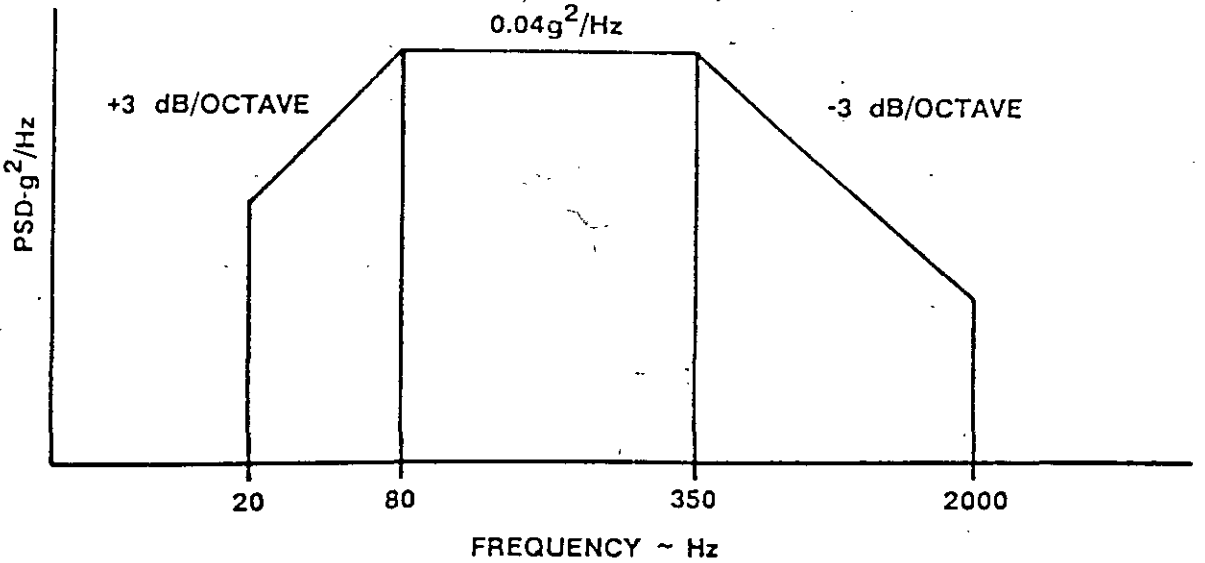


FIGURE 4. Workmanship random vibration.

daily during hot and cold periods, after the production AM-3889s have stabilized at inspection temperatures. The last two temperature cycles shall be failure free. The thermal stress screening inspection chamber's temperature change rate shall be greater than 10°C per minute. The contractor shall be afforded the option of performing the vibration inspection concurrently with the temperature cycling portion of the screening inspection or as a stand-alone inspection.

4.3.1.4 Production reliability acceptance test (PRAT). Following the environmental stress screening inspection of 4.3.1.3 each AM-3889 shall be subjected to a production reliability acceptance test as a means for verifying the MTBF requirement of 3.5.1.3 (1000 hours). Only those AM-3889s that have successfully completed both the environmental stress screening inspection and the production reliability acceptance test shall be delivered to the procuring activity. The production reliability acceptance test shall be in accordance with MIL-STD-781, Test Plan XVIIIC. Each AM-3889 shall accumulate a minimum of 45 operating hours (last three cycles failure free) under environmental cycling conditions as shown on Figure 5. No particular AM-3889 shall be allowed to accumulate more than 90 hours of production reliability acceptance test time.

4.3.1.5 Determination of compliance. The total production reliability acceptance test time and chargeable failures will be plotted on the Continue Test/Reject Criteria Graph of MIL-STD-781, Test Plan XVIIIC, using a lower test MTBF O_1 of 1000 hours. The upper test MTBF O_0 shall be 2000 hours (see Figure 6). If the plot of test failures versus total test time enters the reject region, testing shall be suspended and the procuring activity along with the quality assurance representative shall be notified.

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PRODUCTION RELIABILITY ACCEPTANCE TEST
ENVIRONMENTAL TEMPERATURE CYCLE
AM-3889/AIC-14

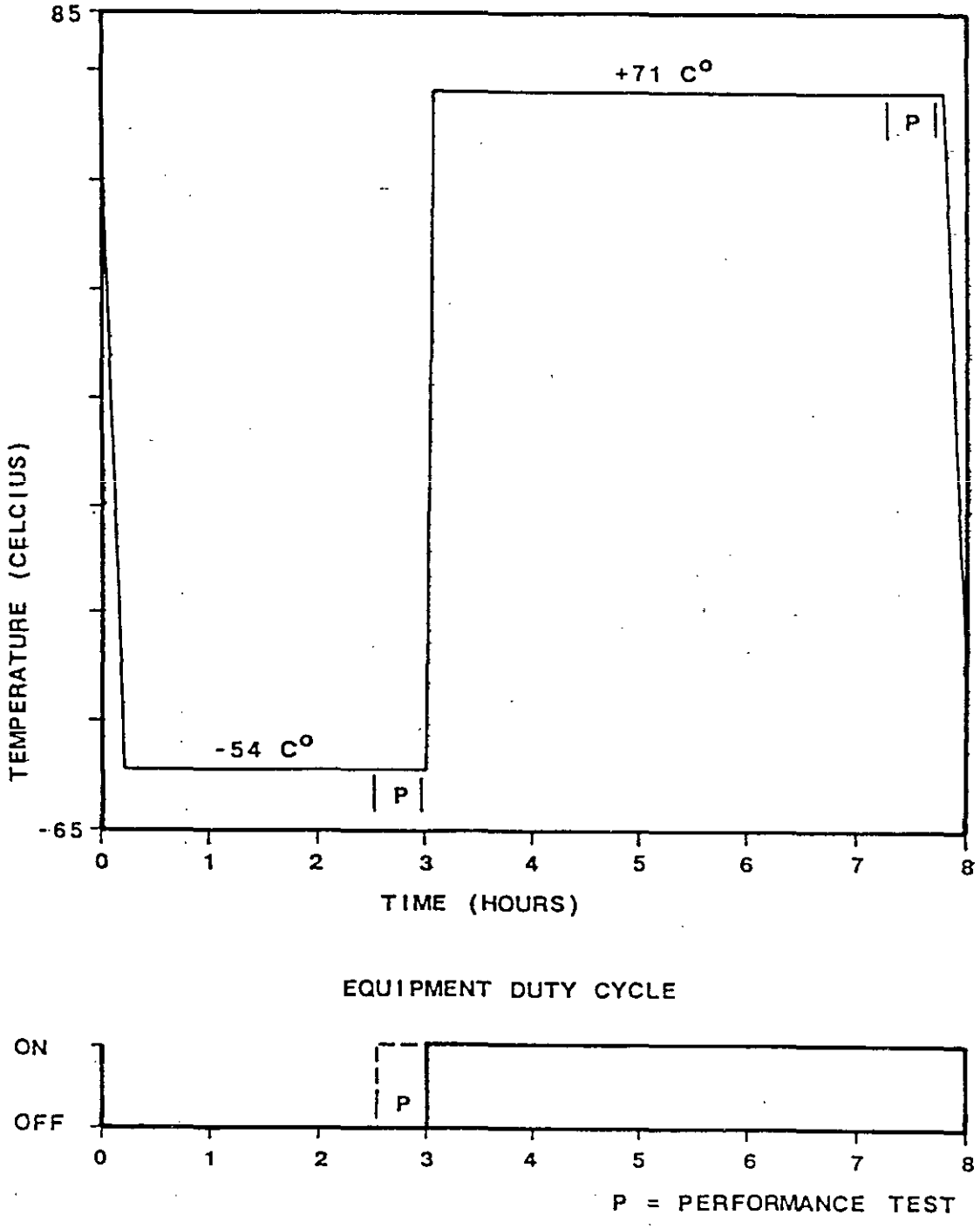


FIGURE 5. Environmental cycling conditions.

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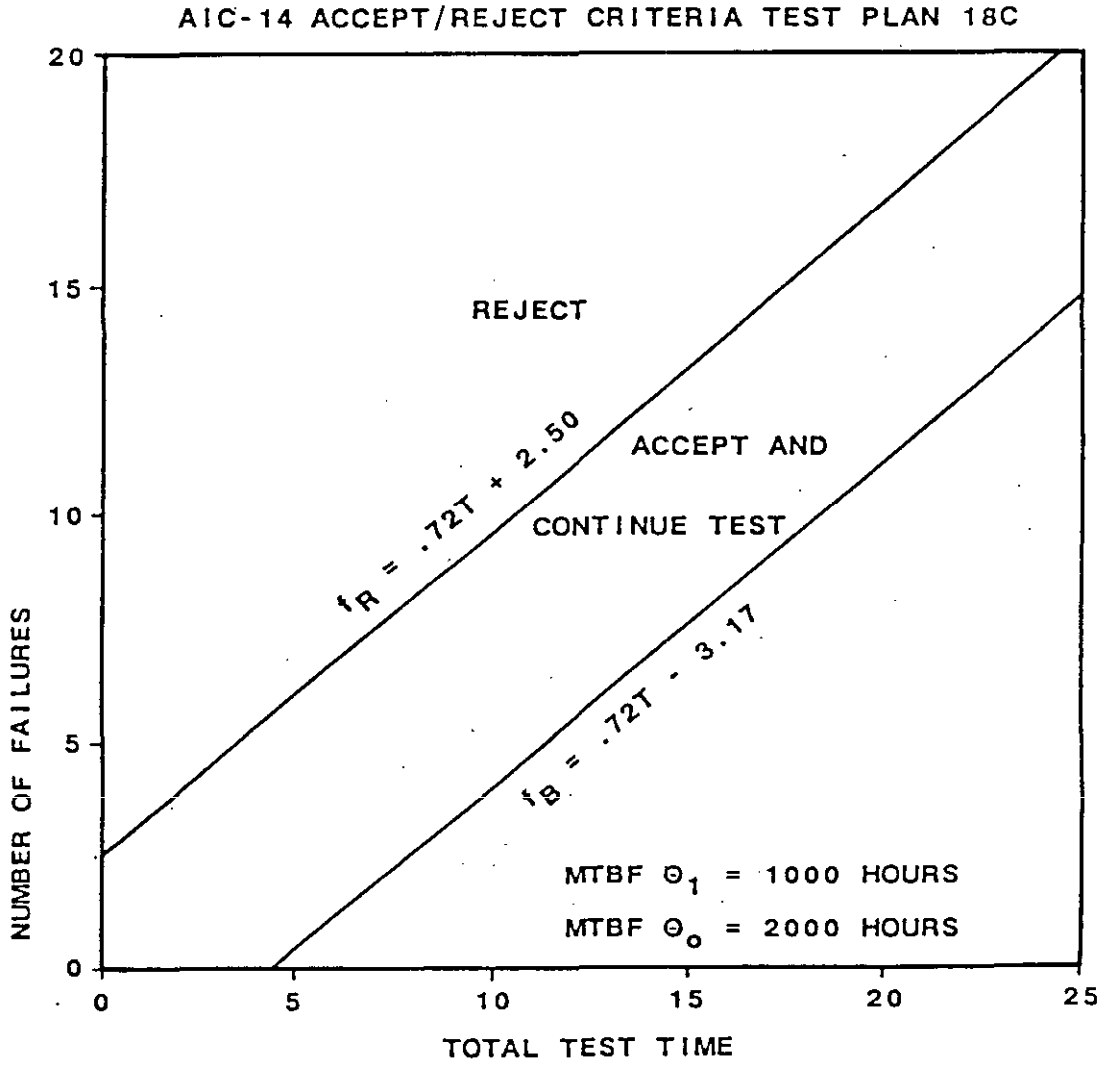


FIGURE 6. Continue test/reject criteria graph.

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4.4 Special inspections. The procuring activity reserves the right to perform any one or group of inspections set forth in this specification when such inspections are deemed necessary by the procuring activity to ensure production item conformance to the specified requirements. Conditions that warrant special inspections shall include:

a. First production AM-3889 incorporating a design or material change.

b. Whenever production is interrupted for more than 1 year.

c. Whenever failure reports or other information indicate that additional inspections are required.

d. Whenever it is deemed necessary to ensure production AM-3889s are equal to the previously approved first article inspection AM-3889 in design, construction, workmanship, performance and quality.

e. As specified in the contract or purchase order (see 6.2.1).

4.5 Classification of failures. Should a failure occur during any inspection, the following action shall be taken:

a. Determine the cause of failure.

b. Determine whether the failure is an isolated case or design or production process defect.

c. Implement corrective action intended to reduce the possibility of the same failure(s) occurring in future inspections.

d. Where practical, include additional sequences during the quality conformance acceptance inspections until assurance is obtained that the defect has been corrected.

4.6 Methods of examination and inspection. All inspections shall be conducted in accordance with Government approved procedures. When approved inspection procedures are available from previous contracts, they may be provided and used upon authorization of the procuring activity. The Government reserves the right to modify procedures or add inspections when it is deemed necessary to ensure AM-3889s are in compliance with this specification and the contract.

5. PACKAGING

5.1 General. Packaging shall be in accordance with the general requirements of MIL-E-17555, supplemented as follows:

5.2 Preservation-packaging. Unless otherwise specified in the contract or purchase order (see 6.2.1) preservation-packaging shall be in accordance with MIL-P-116, Method III.

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5.3 Packing. Unless otherwise specified in the contract or purchase order (see 6.2.1) packing shall be level A. The packaged equipments shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, normal requirements. Sealing (taping) and reinforcing (banding) shall be in accordance with the appendix to PPP-B-636.

5.3.1 Unitized loads. Unitized loads, commensurate with the level of packing specified herein or in the contract (see 6.2.1) shall be used whenever the total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent possible.

5.4 Marking. In addition to any special marking required by the contract (see 6.2.1) each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AM-3889 Audio Frequency Amplifier is intended to be a component of the AN/AIC-14A Intercommunications Set.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of packaging and packing (see Section 5).

c. When a first article is required (see 4.2).

d. If special tests are required (see 4.4).

e. Inspection conditions, if other than specified in 4.1.5.

6.3 First article. When a first article inspection is required, the AM-3889 will be tested and should be a first article sample. The first article should consist of two AM-3889s. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles samples. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to relay on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Subject term (keyword) listing.

Amplifier, Audio Frequency, AM-3889
Control Panel, Master Interphone, C-2645D
Set, Intercommunications, AN/AIC-14A

Preparing Activity:
Navy - AS
(Project 5831-N083)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-55086A (AS)	2. DOCUMENT DATE (YYMMDD) 14 NOVEMBER 1990

3. DOCUMENT TITLE INTERCOMMUNICATIONS SET, AN/AIC-14A AUDIO FREQUENCY AMPLIFIER AM-3889/AIC-14A GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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